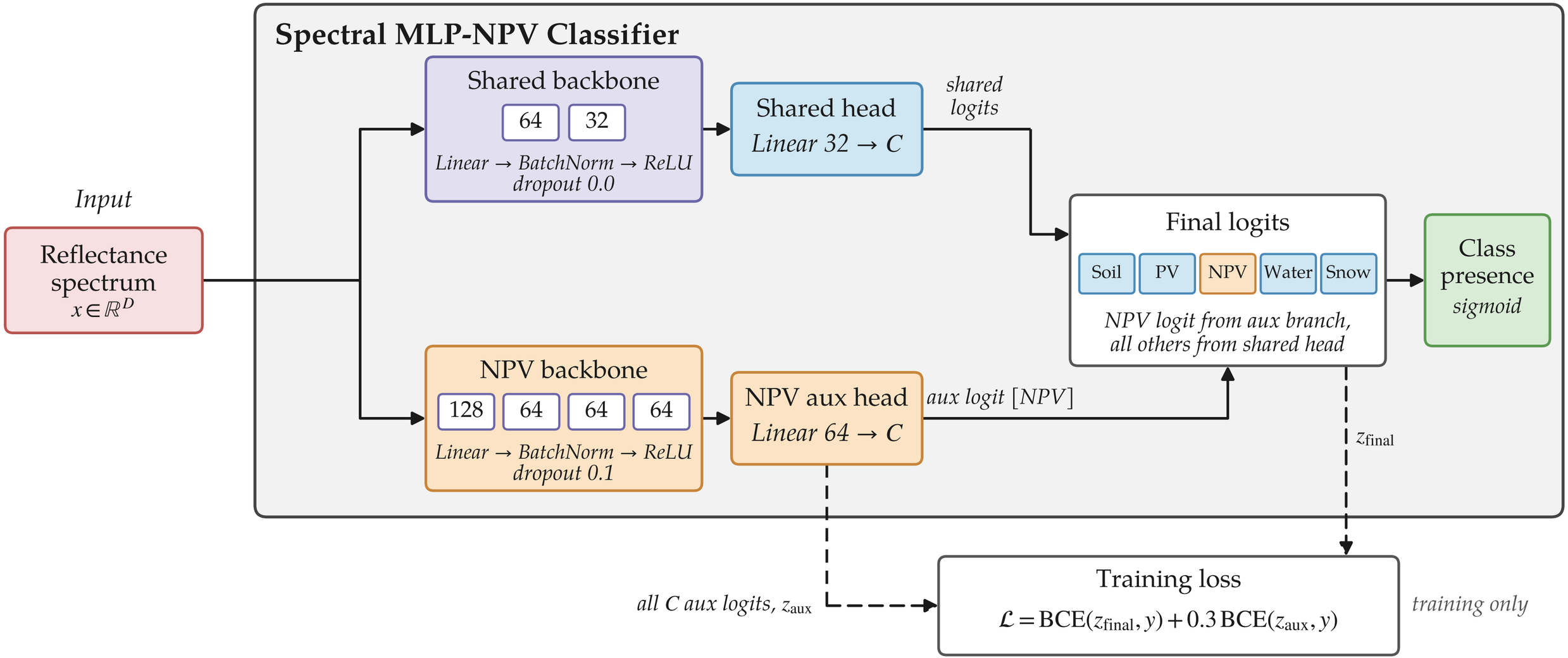

Spectral MLP-NPV Classifier
Input
Reflectance spectrum
$x \in \mathbb{R}^D$
Shared backbone
64
32
Linear → BatchNorm → ReLU
dropout 0.0
Shared head
Linear 32 → C
shared logits
NPV backbone
128
64
64
64
Linear → BatchNorm → ReLU
dropout 0.1
NPV aux head
Linear 64 → C
aux logit [NPV]
Final logits
Soil
PV
NPV
Water
Snow
NPV logit from aux branch,
all others from shared head
Class presence
sigmoid
z_{final}
all C aux logits, z_{aux}
Training loss
$\mathcal{L} = \mathrm{BCE}(z_{final}, y) + 0.3\,\mathrm{BCE}(z_{aux}, y)$
training only